

# Braedyn Thompson

(949) 565-5903 • braedynthompson@berkeley.edu • Berkeley, CA | github.com/braaaceedyn

## EDUCATION

University of California, Berkeley

Graduation: Fall 2027

Major: Computer Science & Data Science - Applied Mathematics & Modeling Emphasis

GPA: 3.75/4.00

Coursework: Machine Learning, Artificial Intelligence, Data Structures, Principles of Data Science, Data Engineering, Data Mining and Analytics, Linear Programming

## WORK & PROFESSIONAL EXPERIENCE

### Lawrence Berkeley National Laboratory

Berkeley, CA

Data Science & Machine Learning Research Intern

Sep 2026 – Present

- Implemented PatchTST (PyTorch, neuralforecast) for multivariate time-series forecasting of XCache traffic signals across multiple horizons, systematically tuning lookback windows and outperforming a persistence baseline on all targets by up to 24% lower RMSE.
- Exposed the ceiling on peak-load predictability and guided feature engineering by building reproducible Pandas EDA pipelines over three full years of traffic logs, visualizing seasonality, change-points, and right-skewed peaks with Matplotlib, Seaborn, and Plotly while applying rolling-origin cross-validation, and performing comprehensive multi-horizon forecasting evaluations across all feature sets successfully.

### California Baptist University

Riverside, CA

Machine Learning Research Intern

May 2026 – Present

- Diagnosed root cause of a stalled inverse-FEA predictor by benchmarking 7 ML models (Random Forest, XGBoost, HGB/ExtraTrees ensembles, PCA pipelines) and running paired clean-vs-noisy signal analysis via a noise-identifiability sweep, proving the  $R^2 \approx 0.10$  ceiling was a data limitation rather than a model limitation, redirecting the team from futile model tuning in preparation for IEEE submission.
- Raised earthquake-detection accuracy to 0.9998 ROC-AUC (95% CI 0.9997–0.9999) vs. 0.816 for STA/LTA baseline, and magnitude  $R^2$  to 0.951 vs. 0.886 for amp-dist baseline, by training PyTorch CNN→Transformer and CNN→GNN→Transformer ensembles on 50K+ waveform windows and 6,243 Southern California quakes, using chronological splits and event-clustered bootstrap confidence intervals.
- Shipped a self-maintaining real-time alert system that confirmed real quakes 93% of the time with zero false alerts on held-out replay days, by building a 19-station SeedLink pipeline: multi-station location with a negative-evidence check, user-selectable two-stage FCM pushes, and an MLOps loop comprising an MLflow registry, statistically gated champion/challenger promotion with a replay acceptance test, Dagster monthly retraining, Evidently station drift monitoring, and GitHub Actions CI/CD deploying to an Oracle Cloud A1 VM.

### NumIsToken

Berkeley, CA

Full-Stack Software Engineer Intern

Sep 2025 – Jun 2026

- Eliminated an entire class of unrecoverable stuck-transaction failures and restored interrupted multi-step workflows by designing and implementing persistent Redemption History and Detail services with robust state management and payment-resumption logic.
- Reduced redemption action load times by ~85% (~2–3s to ~300ms) by removing redundant database lookups and repeated validation checks to streamline backend lifecycle APIs, then containerizing and deploying backend services with Docker for production readiness.

### The Kigumi Group

Hong Kong

Backend Software Engineer Intern

Jun 2025 – Sep 2025

- Increased test coverage from 90% to 98% and resolved recurring AI assistant pipeline failures by deploying targeted debugging with unit and integration tests to fix connection drops and malformed queries for GPT and image models, enhancing reliability and consistency.

## PROJECTS: Portfolio - braedynthompson.com

### Academy of Testers

Berkeley, CA

Machine Learning & Artificial Intelligence Software Developer

Jan 2023 – Jun 2026

- Architected an adaptive learning engine that models per-student skill mastery by combining Bayesian Knowledge Tracing with Item Response Theory, dynamically selecting questions to maximize information gain at the learner's estimated ability, while extending the 8-skill model with exponential forgetting-curve decay and prerequisite-graph penalty propagation to drive a real-time mastery radar.
- Reduced AP essay-grading variance, achieving a ~23% reduction in scoring error and an improved QWK of 0.62 against official College Board scores by designing a RAG pipeline that grounds LLM evaluations in retrieved rubric clauses and score-banded exemplars.

### BearLM

Berkeley, CA

Data & Artificial Intelligence Engineer

Aug 2026 – Present

- Built a fully local, zero-cost RAG Q&A assistant (LangChain, Ollama, Llama 3.1 8B, nomic-embed-text, Chroma) over ~10K chunks spanning UC Berkeley CS/DS courses; blended keyword and meaning-based search (BM25 + vector embeddings) with cross-encoder reranking to pull the most relevant sources, successfully placing the true answer as the top result 82% of the time (up from 56%).
- Reduced hallucinations by grounding responses strictly in retrieved sources, combining a relevance filter that scores and drops weak matches, query rewriting for sharper retrieval, context-only prompting; raising RAGAS faithfulness from 0.55 to 0.83 (0.97 relevance).
- Engineered an ETL pipeline ingesting heterogeneous source files (PDF, Markdown, Jupyter notebooks), token-chunking them with metadata, and batch-embedding ~10K records with retry/backoff into Chroma, BM25 indices, and SQLite for hybrid search & analytics.

## TECHNICAL SKILLS

PyTorch, scikit-learn, XGBoost, Neural Networks, Deep Learning, CNN, Transformers, GNN, Llama, Prompt Engineering, RAG, LangChain, Ollama, Chroma, Vector Search, Hybrid Search, Reranking, RAGAS, Batch Embedding, Data Ingestion, ETL Pipelines, PostgreSQL, MySQL, SQLite, Docker, Git, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Python, SQL, Java, Spring Boot, FastAPI, Vite, Unit & Integration Testing